

# KHALID AHMAD

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## Research Interests

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Applied Machine Learning, Computer Vision, Natural Language Processing, AI4Health

## Education

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Federal University, Lokoja

Sep. 2018 – Oct. 2023

B.Sc. Computer Science CGPA: 4.75/5.0 (3.9/4.0)

Lokoja, Nigeria

## Research Experience

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Federal University, Lokoja

Oct 2022 – Present

Research Assistant

Lokoja, Nigeria — Hybrid

- Conducting a literature review for the development of an AI-based system for early detection of plant diseases, linking plant health to healthcare through food safety and nutrition.

## Publications

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- Khalid, H.A.**, Fati, O.O, Abdulkakeem A.O, & Aifuobhokhan, J. (Manuscript in Progress). PERG-EyeNet: An Interpretable Boosting Framework for Automated Classification of Pattern Electroretinogram Signals with Clinical Metadata.
- Aifuobhokhan, J., **Khalid, H.A.**, Ekechi, C.C., Olanrewaju, A.O., Afuadajo, E., Bowale, D.A., & Inioluwa, O.M. (Under Review). Predicting Thyroid Cancer Recurrence Using Machine Learning: An Artificial Intelligence Approach to Clinical Oncology. *International Journal of Advances in Artificial Intelligence and Machine Learning*.
- Aderemi, I.A., Kehinde, T.O., Ugochukwu, D.O., **Khalid, H.A.**, Adjei, K.Y., & Chijioke, C.E. (2025). Beyond the Black Box: A Systematic Review of Explainable AI for Transparent and Trustworthy Water Quality Monitoring. *IEEE Sensors Reviews*, August 2025. [DOI]
- Khalid, H.A.**, & Fati, O.O. (2025). A real-time road anomaly detection via masked autoencoder-enhanced vision transformers. *International Journal of Innovative Research in Engineering & Management*. [DOI]
- Kossy, S., Saidu, M., **Khalid, H.A.**, Ogeah, P.U., Bakare-Abidola, T., Aanuoluwa, T.I., & Obafemi, A.B. (2025). Advancing environmental sustainability through emerging AI-based monitoring and mitigation strategies for microplastic pollution in aquatic ecosystems. *World Journal of Biology Pharmacy and Health Sciences*. [DOI]

## Conferences & Presentations

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- Khalid, H.A.**, Ahmad, M.B. Efficient Multiclass Classification of Mental Health Conditions in Reddit Posts: A Comparative Study of Transformer Models and Traditional Machine Learning. *National Conference on Smart Technologies for Societal Development (STSD-2025)*, Sept 2025
- Khalid, H.A.**, Fati, O.O., Faruk, A., & Ismail, A. Random forest-based predictive model for estimating first-year GPA in Nigerian tertiary institutions.
- Khalid, H.A.**. Exploring Technology Hotspots: The Impact of Mathematics on Computer Science. *National Association of Computing Students (NACOS) Workshop, FUL Chapter*, Aug 2022
- Khalid, H.A.**, Musa, A., & Grace, A. Four Ways Traffic Light Design and Implementation. *Department of Computer Science Symposium, Federal University Lokoja*, Oct 2021

## Academic & Professional Service

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Deep Learning Indaba

Jul 2025 – Aug 2025

Judge

Kigali, Rwanda — Hybrid

- Selected as a judge for the Deep Learning Indaba 2025 AI Community Challenge in Kigali to support the evaluation of innovative AI projects from over 160 teams across more than 20 countries.
- Supported the advancement of real-world AI impact by promoting emerging research and recognizing talent within the African AI community.

## Technical Skills

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**Languages & Frameworks:** Python, C++, JavaScript, HTML, CSS, SQL, Flask, Bootstrap, Streamlit

**Machine Learning & AI:** PyTorch/TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, NLTK, Matplotlib

**Developer Tools & Other Technologies:** Git, Docker, VS Code, Jupyter, GitHub, Slack, Azure, MySQL, Hugging Face

**Writing Tools:** LaTeX, Microsoft Word, Google Docs.

## Teaching Experience

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### Digital onboarders

Jan 2024 – Nov 2024

CS Mentor & AI Workshop Facilitator

Lafia, Nigeria — Hybrid

- \* Delivered hands-on instruction in Python and core CS subjects to 75+ students, guiding them in developing web applications and mini-projects.
- \* Led 25+ lab sessions using Jupyter Notebooks and GitHub; graded 50+ student projects with personalized feedback to foster learning.
- \* Participated in 3+ national workshops and the Miva Conference on AI, contributing to discussions on AI and Nigeria's digital economy.

### Federal University, Lokoja

Mar 2019 – Jul 2023

Academic Tutor

Lokoja, Nigeria — Onsite

- \* Tutored over 2,500 students in core STEM subjects including Physics, Mathematics, Statistics, and Programming, covering topics like Linear Algebra, Calculus, and Probability.
- \* Led 30+ exam prep sessions and 80+ personalized 1-on-1 tutorials in Python and C++, contributing to a 20-35% documented improvement in student performance.
- \* Assisted 5+ faculty members in teaching CS courses by simplifying complex topics in Operating Systems, Compiler Design, Discrete Structures, and Machine Learning, contributing to a 15% average improvement in student assessment scores.

### Cali Vision Academy

Feb 2021 – Sept 2022

ICT Instructor

Lokoja, Nigeria — Onsite

- \* Taught over 120 secondary school students IT fundamentals, productivity tools, and introductory programming, resulting in 85% of students earning a distinction (A) in WAEC Computer Studies.
- \* Developed and delivered a structured 3-term ICT curriculum with 24 lesson plans and 50+ practical exercises, improving average student performance by 35%.
- \* Organized 2 intra-school ICT competitions that increased STEM club participation by 60% and enhanced student engagement in technology.

## Industrial Experience

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### Omdena

Feb 2025 – May 2025

Machine Learning Engineer

New York, USA — Remote

- \* Collaborated with a global team of 10+ engineers on a Smart Farming project to boost yields for 10,000+ farmers in Kano State, Nigeria.
- \* Cleaned and engineered features from 148,666 soil moisture records to enhance model inputs, improving data quality and system reliability.
- \* Trained a Random Forest model that achieved 99% precision in classifying plant health indicators, improving detection accuracy by 9% over baseline.

### Saint Louis University

Jun 2023 – Jul 2023

Data Analyst

Missouri, USA — Remote

- \* Analyzed digital marketing campaign data for Finance and Marketing teams, tracking CPC, CPR, reach, and engagement metrics across 10 campaigns.
- \* Built interactive dashboards using Python, SQL, and Power BI to visualize performance insights, identifying top-performing campaigns (2 and 8).
- \* Reported ad spend of INR 12,088.61, reaching 188,868 users and generating 5,257 unique link clicks; recommended optimizations for underperformers (3 and 10).

### Nigerian Midstream & Downstream Petroleum Regulatory Authority

Mar 2022 – Aug 2022

Data Scientist Intern

Lokoja, Nigeria — Onsite

- \* Analyzed 150+ petrol samples from 40+ Lokoja filling stations, uncovering a 22% adulteration rate and informing regulatory action.
- \* Built ML models (Logistic Regression, Decision Trees, Random Forest, SVM) with 75% accuracy in detecting fuel adulteration using preprocessed data in Python.
- \* Reduced manual inspection workload by 30% by integrating predictive analytics into existing regulatory workflows.

## Leadership & Service

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### Senator Representative

Federal University Lokoja

National Association of Lokoja Local Government Students (NALLGS)

May 2020 – Jun 2022

- \* Represented interests of over 50 students from Lokoja Local Government at national academic forums, influencing key policy discussions.
- \* Facilitated communication between 50+ students and association leadership, ensuring timely resolution of student concerns.

## General Secretary

NACOS Electoral Committee

*Nigeria Association Of Computing Students.*

*June 2022*

- \* Coordinated the organization and execution of elections involving 5,000+ student voters, increasing participation by 15%.
- \* Ensured transparency and fairness by maintaining accurate records and overseeing electoral procedures, reducing disputes by 30%.

## Honors & Awards

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- \* Wale Lab Research Internship – 2025: Fully funded AI healthcare research internship.
- \* Departmental and Faculty Fee Waiver – 2023: Awarded for top academic performance and engagement.
- \* Ayomide Ajiboye Scholar (USA) – 2023: Scholarship and mentorship for academic excellence.
- \* Student Tutor of the Year – 2023: Recognized for outstanding peer tutoring.
- \* Department Contribution Award – 2022: Honored for impactful research and academic service.
- \* NMDPRA Recommendation – 2022: Commended for exceptional work with Nigeria's petroleum authority.
- \* Dean's Honor Award – 2020–2023: Multiple-year award for sustained academic excellence.
- \* Kogi State Bursary Scholar – 2020: Financial award for academic merit and need.

## Volunteering Experience

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### Digital Onboarders (DODEEL)

**Feb 2024 – Aug 2024**

*Community Development Service (CDS) Volunteer*

*Lafia, Nigeria — Onsite*

- Led digital literacy sensitization programs in 5+ Government Secondary Schools in Lafia, teaching foundational computer skills to over 300 students.
- Mentored 50+ students in Lafia on personal development and digital skills awareness.
- Trained 40+ small business owners in Lafia on using Google My Business and Maps to improve their digital visibility.

### National Youth Service Corps (NYSC)

**Nov 2023 – Dec 2023**

*Data Entry Analyst*

*Edo, Nigeria — Onsite*

- Processed and verified records for over 3,500 NYSC Batch C2 corps members deployed across Edo State.
- Conducted exploratory data analysis (EDA) to extract demographic insights on gender distribution, educational background, and skill acquisition program participation, aiding program planning and reporting.

### United Nations International Children's Emergency Fund (UNICEF)

**Aug 2023 – Dec 2023**

*Data Entry Analyst*

*Lokoja, Nigeria — Hybrid*

- Collected data from over 500 households, 3 hospitals, and 2 government agencies in Lokoja to support accurate population health reporting.
- Registered more than 1,200 newborns into UNICEF's database and facilitated the distribution of 1,200+ birth certificates to local healthcare facilities.

### Federal University Lokoja

**Mar 2021 – Oct 2023**

*Social Media Moderator*

*Lokoja, Nigeria — Hybrid*

- Co-founded and managed a student-led social media support group with 80,000+ members at Federal University Lokoja, dedicated to assisting new and returning students with academic, housing, and campus-related inquiries.
- Moderated discussions and responded to hundreds of student inquiries weekly, fostering a trusted and inclusive digital community.
- Developed and shared informational content that improved student onboarding, reduced confusion during registration periods, and boosted engagement across the platform.

## Professional Affiliations

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- Member, Black in AI (BAI)
- Member, Data Science Nigeria (DSN)
- Member, National Association of Computing Students (NACOS)
- Member, Nigeria Computer Society (NCS)
- Core Member, Google Developer Student Clubs (GDSC)